

Zixuan Liu

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Research Interests

My research focuses on developing intelligent surgical assistive systems that integrate mixed reality (MR), robotics, computer vision, and deformable registration to enhance the efficiency of minimally invasive surgeries. I am particularly interested in (1) utilizing MR and computer vision algorithms to provide intraoperative guidance and navigation, (2) developing robotic systems with intelligent decision strategies for understanding the surgical scene, and (3) deformable modeling for soft tissue surgery, especially for breast-conserving surgery (BCS).

Education

PhD	Vanderbilt University , Computer Science	Nashville, TN, USA Aug 2024 – present
	- Current GPA: 4.0/4.0 - Advisors: Michael I. Miga (primary), Jie Ying Wu (co-advisor) - Affiliations: Biomedical Modeling Laboratory (BML); Machine Automation, Perception, and Learning (MAPLE) Lab; Vanderbilt Institute for Surgery and Engineering (VISE) - Research keywords: Computer Aided Intervention, Image-guided Surgery, Surgical Robotics, Mixed Reality	
MSE	Johns Hopkins University , Computer Science	Baltimore, MD, USA Aug 2022 – May 2024
	- GPA: 3.79/4.0 - Affiliation: Laboratory for Computational Sensing and Robotics (LCSR)	
BS	Wenzhou-Kean University , Computer Science (minor in Mathematical Science)	Wenzhou, China Aug 2018 – June 2022
	- Magna Cum Laude - GPA: 3.818/4.0	

Professional Employment

Euclid Techlabs LLC , Robot Algorithm Engineer Intern	Hybrid, Beltsville, MD, USA July 2023 – Sept 2023
Involved in development of an HDR Electronic Brachytherapy system prototype.	

Selected Awards

- Finalist, Best Paper Award, IPCAI 2024 (for [J-3])
- Potential Impact Award, Introduction to Augmented Reality, Johns Hopkins University, 2023
- Best Project Award, Computer Integrated Surgery II, Johns Hopkins University, 2023
- Dean's Scholarship – Research and Innovation, Wenzhou-Kean University, 2021
- Dean's Scholarship – Second Class, Wenzhou-Kean University, 2021
- Dean's Scholarship – Third Class, Wenzhou-Kean University, 2020
- Magna Cum Laude, B.S. in Computer Science, Wenzhou-Kean University, 2022

Journal Articles

ReSNA: A Robotic-Enabled Surgical Navigation Assistant for Breast-Conserving Surgery	2026
Z. Liu, B. Xiang, J. Heiselman, J.Y. Wu, I. Meszoely, M.I. Miga	
10.1109/LRA.2026.3710046 (IEEE Robotics and Automation Letters (RA-L), vol. 11, no. 8, pp. 9994–10001)	

- Stand in Surgeon's Shoes: Virtual Reality Cross-training to Enhance Teamwork in Surgery** 2024
 Finalist for Best Paper Award at IPCAI 2024.
 B.D. Killeen, H. Zhang, L.J. Wang, **Z. Liu**, C. Kleinbeck, M. Rosen, R.H. Taylor, G. Osgood, M. Unberath
[10.1007/s11548-024-03138-7](https://doi.org/10.1007/s11548-024-03138-7) (International Journal of Computer Assisted Radiology and Surgery (IPCAI 2024 Special Issue))
- A Magnetic Resonance Conditional Robot for Lumbar Spinal Injection: Development and Preliminary Validation** 2023
 D. Liu, G. Li, S. Wang, **Z. Liu**, Y. Wang, L. Connolly, G. Shen, K. Cleary, I. Iordachita
[10.1002/rcs.2618](https://doi.org/10.1002/rcs.2618) (International Journal of Medical Robotics and Computer Assisted Surgery (IJMRCARS))
- Optimal Recycle Price Game Theory Model for Second-hand Mobile Phone Recycling** 2021
 K.E. Ehimwenma, S. Krishnamoorthy, **Z. Liu**, Y. Qiu, Y. Liu, W. Dou
[10.1007/s11356-021-17061-w](https://doi.org/10.1007/s11356-021-17061-w) (Environmental Science and Pollution Research International (ESPR))

Conference Papers

- Robotic Stereo-camera System with SAM 2 for Multiview Surface Reconstruction of Anatomical Structures in Surgery** 2026
 Selected for oral presentation.
 B. Xiang, **Z. Liu**, M.I. Miga
[10.1117/12.3087701](https://doi.org/10.1117/12.3087701) (SPIE Medical Imaging 2026: Image-Guided Procedures, Robotic Interventions, and Modeling, vol. 13927, p. 139270P)
- Real-time Optimal Viewpoint Control for Image-guided Surgery using Learning-based Active Perception** 2026
 U. Rajamani, **Z. Liu**, I.M. Meszoely, M.I. Miga
[10.1117/12.3088072](https://doi.org/10.1117/12.3088072) (SPIE Medical Imaging 2026: Image-Guided Procedures, Robotic Interventions, and Modeling, vol. 13927, p. 1392721)
- Augmented Reality-Based Guidance with Deformable Registration in Head and Neck Tumor Resection** 2025
 Q. Yang, F. Li, J. Xu, **Z. Liu**, S. Sridhar, W. Jin, J. Du, J. Heiselman, M. Miga, M. Topf, J.Y. Wu
[10.1007/978-3-032-05114-1_5](https://doi.org/10.1007/978-3-032-05114-1_5) (Medical Image Computing and Computer-Assisted Intervention (MICCAI))
- Development of an MRI Guided Auxiliary Robot for Spinal Injections** 2023
 D. Liu, **Z. Liu**, G. Li, L. Connolly, G. Shen, I. Iordachita
ieeexplore.ieee.org/document/10473547 (IEEE International Conference on Robotic Computing (IRC))

Book Chapters

- Integration of IoT and Big Data in the Field of Entertainment for Recommendation System** 2020
 Y. Liu, W. Dou, **Z. Liu**, E. Szarek, S. Krishnamoorthy
[10.1201/9781003009092-6](https://doi.org/10.1201/9781003009092-6) (Securing IoT and Big Data, pp. 109–121, CRC Press)

Preprints / Under Review

X-GuideAR: An Augmented Reality Framework to Mitigate Radiation Exposure during Fluoroscopic Guidance. A Use Case in S2AI Screw Placement

Under Review.

M. Liu, **Z. Liu**, R. Cai, Y.-C. Ku, S. Gu, A. Jain, A. Martin-Gomez, M. Armand

Invited Talks and Demos

- 2026 International Symposium on Medical Robotics (ISMR), Late-Breaking Research Poster, Knoxville, USA, 04/2026 — "ReSNA: A Robotic-Enabled Surgical Navigation Assistant for Breast-Conserving Surgery."
- 14th VISE Symposium, Oral Presentation, Vanderbilt University, USA, 12/2025.
- 13th VISE Symposium, Vanderbilt University, USA, 12/2024.
- 1st Computer Science Symposium, Wenzhou-Kean University, China, 12/2024.

Teaching

Wenzhou-Kean University, Peer Tutor, Student Academic Support Service Center

2019 – 2021

Service

- Course Assistant, Computer Integrated Surgery I (EN.601.455/655), Johns Hopkins University, Fall 2023
- Student Reviewer, MSE Computer Science Admissions Committee, Johns Hopkins University, Fall 2023 & Fall 2024
- Journal Reviewer, IEEE Transactions on Medical Robotics and Bionics